

Topological Phase Cosmology: Spontaneous Geometric Symmetry Breaking and the Empirical Mapping of Extra-Dimensional Resonances

Xavier Callens¹

¹*SocrateAI Open Lab, Independent Research Node, France*

(Dated: July 14, 2026)

Historical attempts to describe cosmology through transfinite geometry have frequently foundered on non-dynamical numerology. However, the profound underlying intuition—that macroscopic gravitational phenomena are manifestations of shifting topological phases—finds exact realization in modern String Phenomenology. We present Topological Phase Cosmology, a dynamically evolving global theory of the Dark Sector based on Spontaneous Geometric Symmetry Breaking. By coupling the local stress-energy tensor to the Picard-Fuchs moduli space of a K3-fibered Calabi-Yau compactification, we demonstrate how baryonic overdensities trigger continuous topological phase transitions between the Order-3 K3 surface ($S_{1,2}$) and Order-2 elliptic ($S_{2,1}$) geometries. This framework is observationally anchored by a converged GPU tensor analysis of 327,918 SDSS DR17 galaxies performed by the companion **DarkMatterK3** pipeline (analysis artifacts external to this repository; see §??), which reports a macroscopic perturbative bound of $S_{1,2} \leq 1.177$ and isolates a 22-node supercluster intersection peaking at an asymmetry of $\Delta = 47.0$. We frame these as pipeline-reported consistency statements, not independent empirical proofs.

I. INTRODUCTION

To elevate topological intuition into a sound scientific proposal, we must replace static numerology with continuous moduli dynamics. In Topological Phase Cosmology, the dimensionality and symmetry of the internal K3 compactification are dynamical fields. The asymmetry between competing topological solutions acts as an order parameter for the vacuum, directly linking microscopic string theory to macroscopic cosmology.

II. THE GLOBAL THEORY

A. The Swampland Distance Conjecture

In modern string theory, "infinity" at the quantum boundary is rigorously defined by the Swampland Distance Conjecture (SDC). As a scalar field moves an infinite distance in topological parameter space, an infinite tower of stringy states becomes exponentially light:

$$m(\Delta S) \sim m_0 e^{-\alpha \Delta S} \quad (1)$$

Thus, the "infinity" represents an actual physical tower of Kaluza-Klein states that dictate the physics of the dark sector.

B. Dark Matter: Localized Symmetry Breaking

In the isotropic early universe, the geometry possessed unbroken symmetry. As matter decoupled and began to cluster, the local baryonic stress-energy (ρ_b) exerted extreme pressure on the extra dimensions, driving a localized, spontaneous breaking of the geometric symmetry. This forces a transition from a symmetric Order-4

Calabi-Yau phase to the asymmetric $S_{1,2}$ (K3, Order-3) and $S_{2,1}$ (Elliptic, Order-2) phases.

This topological asymmetry acts as a macroscopic order parameter:

$$\Delta(\vec{x}, t) = |S_{1,2}(\vec{x}, t) - S_{2,1}(\vec{x}, t)| > 0. \quad (2)$$

What astrophysicists observe as a "Dark Matter Halo" is the geometric lensing footprint of this localized topological asymmetry.

This picture is empirically supported (not proven) by the companion **DarkMatterK3** GPU pipeline. Its analysis of 327,918 galaxies from the SDSS DR17 catalog reports a macroscopic upper bound of $S_{1,2} \leq 1.177$, which — if the pipeline's operational definition of the $S_{1,2}$ statistic faithfully tracks the theoretical order parameter — confines geometric warping to a perturbative regime in the local universe. The pipeline further isolated, without target preselection, a 22-node supercluster filament intersection at RA 205.0°, Dec +35.0°, characterized by a localized asymmetry spike of $\Delta = 47.0$ (designated K3-DISC-0003). Provenance and interpretation limits for all four numbers are stated in §??.

C. Dark Energy: The Global Vacuum

If Dark Matter is the local excitation of Δ , Dark Energy is driven by global topological invariants. The Hirzebruch signature ($\tau = -16$) of the K3 surface enforces a severe chiral asymmetry via the Atiyah-Singer Index Theorem. This spectral asymmetry prevents the perfect cancellation of vacuum fluctuations, sourcing the Trace Anomaly of the stress-energy tensor ($\langle T^\mu_\mu \rangle \neq 0$). This irreducible vacuum polarization represents the rigorously derived, dynamic quantum origin of the macroscopic expansion force.

III. THE SOCRATEAI SCIENTIFIC AGORA

The transition from a theoretical framework to an established physical paradigm requires massive computational validation. The **SocrateAI Scientific Agora** operates three distinct research pillars:

1. **Formal Verification:** The Swampland Distance bounds, LVS Hessian stability, and exact-rational sieve extractions are machine-certified in the Lean 4 theorem prover.
2. **LSS GPU Tensor Analytics:** The pipeline autonomously projects SDSS/Euclid galaxies into topological tensor grids.
3. **Observational Cross-Validation:** We issue an open call to cross-reference our top geometric nodes ($\Delta = 47.0$) with Weak Lensing shear (κ) maps to *test* the spatial alignment between geometric symmetry breaking and physical dark matter halos — a genuine falsification opportunity: non-alignment would directly undermine the order-parameter interpretation of Δ .

IV. EMPIRICAL STATUS, PROVENANCE, AND PHYSICAL INTERPRETATION

A. Provenance of the quoted statistics

The four headline numbers in this paper — the 327,918-galaxy sample, the bound $S_{1,2} \leq 1.177$, the phase-transition threshold $S_{1,2} \geq 1.8$, and the K3-DISC-0003 spike $\Delta = 47.0$ — are outputs of the companion **DarkMatterK3 / K3@Home GPU tensor pipeline**. The committed code and data artifacts of the present repository contain the pipeline’s *algorithmic components* (topological FFT, Poisson null-hypothesis harness, tensor-grid projection in `lss_tensor_analytics/`) and a 500-galaxy live-query KK projection (`data/ws9_sdss_kk_map.csv`), but not the full converged run behind these four numbers. Until that run is archived with a reproduction script, these values carry the epistemic status of *reported pipeline outputs*, and every interpretive statement below is conditional on them. The threshold 1.8 is additionally a model-internal definition (the value of the order parameter at which the perturbative expansion of the phase-transition free energy is deemed to break down), not an independently derived critical point.

B. Freeze-out: the reconciliation with the 5D resonance framework

Lee & Tsai’s orbifold model requires dark matter that interacted strongly in the dense early universe but is nearly inert today. Within our framework the two constructions occupy complementary roles: the $S^1/(Z_2 \times Z'_2)$

orbifold supplies the *microscopic* mass-generation mechanism (KK quantization against the compactification radius R), while the TDA pipeline supplies the *macroscopic* observable (the density-coupled order parameter Δ , with the local baryonic field “pinching” the effective R). No browser-side simulation of the orbifold itself is required — or possible — in this division of labor: the orbifold physics enters the observatory only through the parameterized response $m_{\text{eff}}(\Delta)$, and the citizen-science pipeline’s task is to measure the macroscopic statistic, not to solve the 5D field equations. Under this reading, the reported bound $S_{1,2} \leq 1.177$ is *consistent with* — though it does not by itself prove — the frozen-out phase the 5D framework requires at $z \approx 0$: a locally active resonance would have manifested as sectors approaching or exceeding the transition threshold.

C. Sector-level stability of the local vacuum

The pipeline’s sector decomposition reports $S_{1,2} < 1.3$ across all analyzed sky sectors, uniformly below the 1.8 transition threshold. The physically expected reading is that galaxy groups and satellite systems are *macroscopically non-resonant*: a full topological phase transition sustained across a ~ 10 -degree sector of the local volume would correspond to a non-perturbative geometric instability, which manifestly does not occur. The margin between 1.177 and 1.8 (a $\sim 17.7\%$ maximal warping relative to the symmetric point, against a $\sim 80\%$ transition criterion) quantifies — conditionally on the pipeline definitions above — the stability of the late-time ($z < 0.5$) vacuum, and relegates extreme resonances to micro-environments (early-universe direct collapse, black-hole horizons) exactly as the See-Saw picture requires. We stress the epistemic asymmetry: a sector *violating* the bound would have been a striking positive signal; its absence is a consistency check, not a discovery.

D. The T^2 proxy versus Poisson mocks: an algorithmic, not cosmological, victory

In null-hypothesis runs (cf. the in-repository harness `lss_tensor_analytics/null_hypothesis_test.py`), shuffled Poisson mock catalogs yield a *higher* T^2 topological proxy than the real galaxy data. This direction is informative: uniform noise fragments into many small spurious components and noisy tunnels (inflating the β_0/β_1 -driven proxy), whereas gravitationally organized matter concentrates into fewer, smoother, connected filaments and coherent voids. The real data scoring *below* the mocks therefore indicates the pipeline is isolating organized large-scale structure rather than hallucinating topology from noise. We frame this as a validation of the *algorithm’s discriminating power* — a necessary sanity condition — and explicitly not as a measurement of extra-dimensional physics: any

topology-sensitive statistic applied to a gravitationally clustered field versus a Poisson field is expected to show this asymmetry, independently of the K3 interpretation.

V. CONCLUSION

By discarding unsupported numerology in favor of Spontaneous Geometric Symmetry Breaking, the

Swampland Distance Conjecture, and the Trace Anomaly, we have formulated a falsifiable global framework for the Dark Sector, observationally anchored by SDSS GPU tensor analytics (with the provenance qualifications of §??) and formally scaffolded by Lean 4 kernel verification of its exact-arithmetic core. Its promotion from framework to established theory rests on the open tests enumerated above — most sharply, the weak-lensing cross-match of the reported $\Delta = 47.0$ node.